ZCAS UNIVERSITY

CCS4621 - ADVANCED NETWORKING

END OF SEMESTER 1 EXAMINATION

3RD JUNE, 2026

14:00 HRS-17:00HRS

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

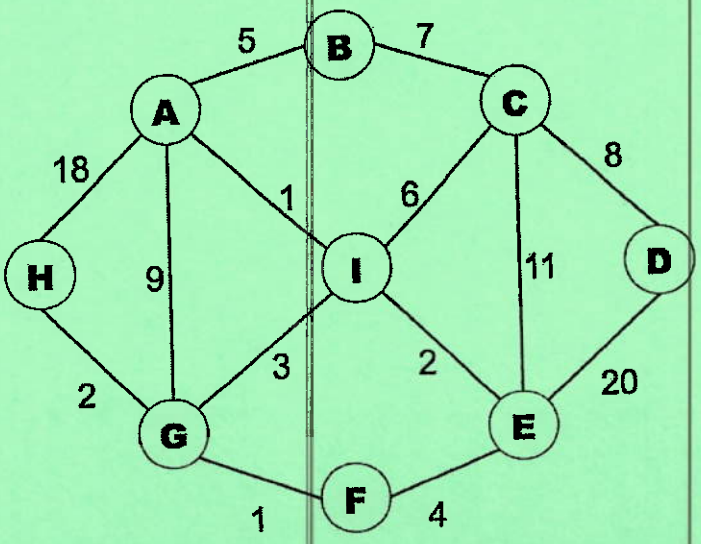
INSTRUCTIONS:

1. Section A: This section is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

(a) Consider the weighted graph below with nodes A, B, C, D, E, F, G, H, I.



(a) Compute the adjacency matrix and the incidence matrix of the network structure. **[10 Marks]**

(b) Using Dijkstra's Algorithm, determine the shortest path from node H to node B. Clearly show all iterations (distance table updates). **[10 Marks]**

(c) Briefly explain why Dijkstra's Algorithm does not work correctly with negative edge weights. **[5 Marks]**

(d) Does the graph above exhibit Euler path and or Euler circuit? Explain your answer using vertex degrees. **[5 Marks]**

(e) AS100 runs OSPF internally with the link costs shown in the topology above. Router A peers with AS200 and receives a BGP UPDATE for prefix 172.16.0.0/16 with AS-PATH [200, 300] and LOCAL-PREF 150. Router E peers with AS400 and receives a BGP UPDATE for the same prefix with AS-PATH [400] and LOCAL-PREF 100.

Apply the BGP decision process in the correct order to identify the preferred exit router, then trace the full OSPF path from router H to that exit router using the link costs shown. **[10 Marks]**

(Total: 40 marks)

SECTION B: Attempt any THREE questions. Each questions carries 20 marks.

Question 2

Consider the network structure shown below in Figure 3 and answer the questions that follow.

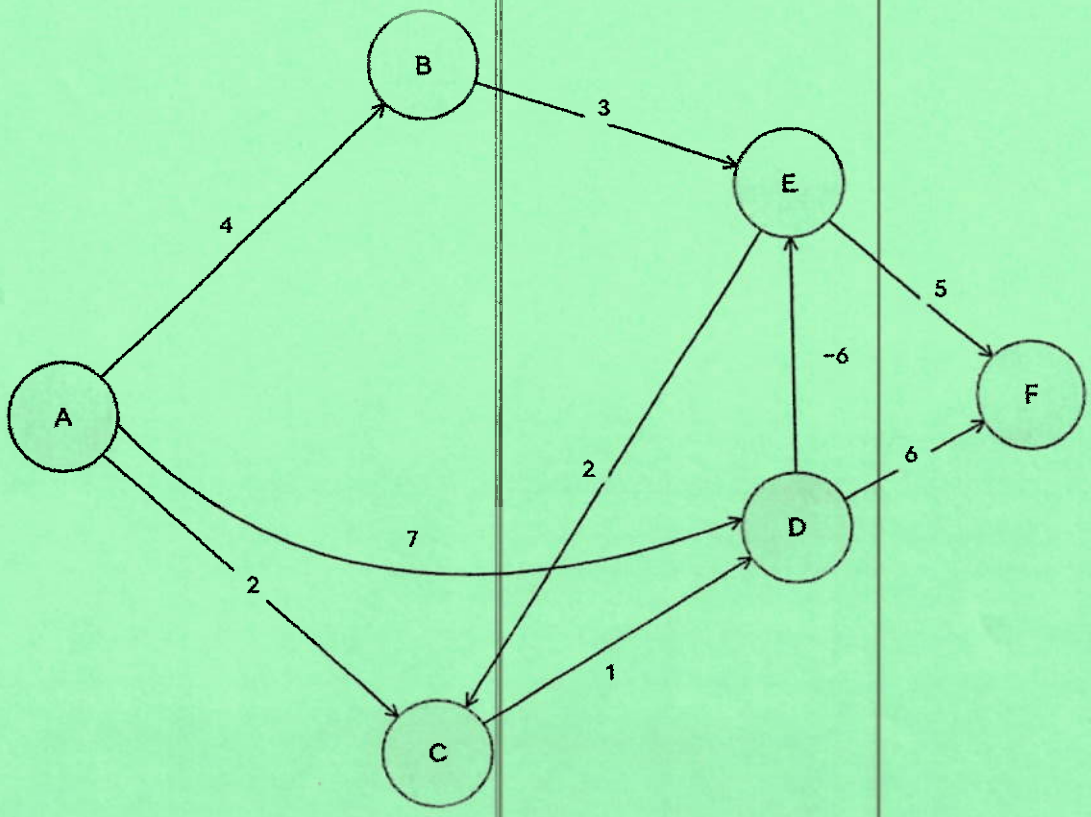


Fig.3 Switched network.

(a) Compute the binary adjacency matrix of the graph. **[5 Marks]**

(b) Compute the incidence matrix of the graph. **[5 Marks]**

(c) Using this graph, apply the Bellman-Ford algorithm with source vertex A, showing your distance table after each relaxation pass. Hence determine whether a shortest path from A to F exists, justifying your answer with reference to the structure of the graph, the behaviour of the algorithm across iterations, and the theoretical condition under which shortest-path problems of this nature have no solution. **[10 Marks]**

(20 Marks)

Question 3

WiFi technology plays a crucial role in modern networking, providing the backbone for a wide range of wireless communications. It has become an essential component in both personal and professional environments. Carefully, consider Table 1 below and answer the questions that follow regarding WiFi technologies.

Table 1: WiFi Technology Characteristics.

WiFi Version	IEEE Standard	Frequency	Max Speed	Modulation	Security
WiFi 1					
WiFi 2					
WiFi 6					
WiFi 7					

a) Fill in and complete Table 1 depicting the various characteristics of WiFi technologies. **[10 Marks]**

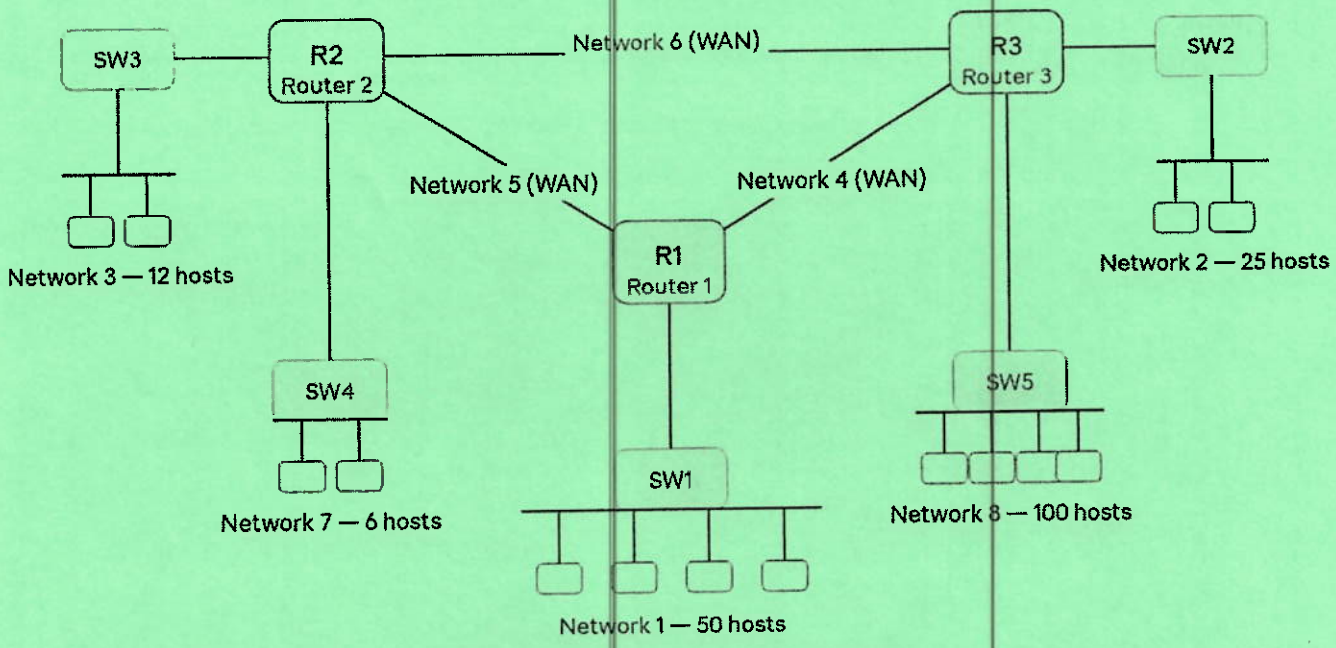
b) With the help of RTS/CTS, explain the CSMA/CA media access mechanism. **[5 Marks]**

c) Discuss any 5 latest technology developments introduced in the latest WiFi version. **[5 Marks]**

(20 Marks)

Question 4

As a network administrator, you have been allocated the address block 193.165.10.0/24 and you must design a VLSM addressing scheme for the network topology shown below. The network consists of three routers (R1, R2, R3) interconnected by point-to-point WAN links, each serving a number of LAN segments with the following host requirements. The routers are all modern and support all subnet mask types.



(a) Using VLSM, allocate subnets from the 192.168.10.0/24 address block to satisfy each network's host requirement with minimum address wastage. Present your answers in the table below in the exact order of networks.

	Network ID	Subnet Mask	CIDR Netmask	Broadcast ID	Wildcard Mask

Network 8					
Network 1					
Network 2					
Network 3					
Network 7					
Network 6					
Network 5					
Network 4					

[20 Marks]

Question 5

A large multinational enterprise operates a core network spanning three continents with six backbone routers: R1 (Lusaka), R2 (Johannesburg), R3 (London), R4 (Frankfurt), R5 (Dubai), and R6 (Singapore). R1 and R2 are Label Edge Routers (LERs); R3, R4, R5 are Label Switch Routers (LSRs); and R6 is both an LER and an SDN-enabled OpenFlow node. The SDN controller manages all flow tables on R6 and its terminating links. All routers operate in IPv6-only mode on inter-router links, and MPLS with LDP runs across the entire backbone.

(a) An IPv6 packet with source address 2001:db8:1::A/48 and destination 2001:db8:6::F/48 enters at R1. Describe in detail its end-to-end journey through the MPLS backbone, explaining label imposition at R1, label swapping at each LSR, and label disposition at R6. Clearly distinguish between the Forwarding Equivalence Class (FEC), the label stack structure, and how the TTL field is processed and propagated at each hop across the IPv6-MPLS boundary. **[6 Marks]**

(b) The SDN controller receives a request to establish a Label Switched Path from R1 to R6 subject to the following constraints: maximum end-to-end delay of 35ms, minimum bandwidth of 500Mbps, and a regulatory compliance policy prohibiting simultaneous traversal of both R4 and R5. Using RSVP-TE, detail the PATH and RESV message exchange, the construction and interpretation of Explicit Route Objects (EROs), and precisely how the SDN controller enforces the compliance constraint by programming R6's OpenFlow flow table. **[6 Marks]**

(c) Two failures occur simultaneously: the R3–R4 link fails, and the SDN controller loses connectivity to R6 for 8 seconds before recovering. For the link failure, explain how MPLS Fast Reroute (FRR) responds, distinguishing between link protection and node protection mechanisms and the role of bypass LSPs. For the controller outage, explain R6's behaviour under OpenFlow fail-safe mode, whether existing flow table entries remain valid, and identify the precise conditions under which IPv6 packets would be permanently lost during either event. **[5 Marks]**

(d) The network engineer proposes replacing LDP with Segment Routing over IPv6 (SRv6) to eliminate the separate MPLS control plane. Critically evaluate this against the existing architecture across four dimensions: control plane complexity, interoperability with the SDN controller, label space scalability, and traffic engineering capability. Conclude with a technically justified recommendation on whether the migration should proceed. **[3 Marks]**

(20 Marks)

---------- **END OF EXAMINATION** ----------